

Al-Aqsa University
Faculty of Computers and IT
Dept. of Info & Computer science



جامعة الأقصى
كلية الحاسبات وتكنولوجيا المعلومات
قسم علم الحاسوب والمعلومات

Information Retrieval

Final Report – TF-IDF Search Engine

Presented to: **Eng. Mohammed Altahrawi**

Student name: **Deema Mohammed AL-Maqadma**
Student number: **2320230766**

Semester: First Semester 2025/2026

1. Introduction

This project implements a simple text-based search engine using TF-IDF and Cosine Similarity.

The goal is to understand how Information Retrieval (IR) systems process documents, represent them numerically, and rank them based on relevance to a user query.

The entire work was completed in Google Colab, following the same structure demonstrated by the instructor.

2. Dataset

I prepared a dataset of 20 external text documents, each saved as a .txt file. Each document contains a short paragraph about a different topic such as AI, cybersecurity, renewable energy, education, robotics, psychology, marketing, and more.

All files were placed inside a folder named documents, then uploaded to Google Colab.

The code automatically read all files, stored their content in a list called documents, and stored filenames for ranking output.

This simulates a real-world IR system that works on external document collections.

3. Preprocessing

Before applying TF-IDF, I cleaned all documents using the following steps:

1. Lowercasing – to unify text.
2. Removing punctuation – using string.punctuation.
3. Tokenization – splitting text into words.
4. Removing English stopwords – using ENGLISHSTOPWORDS.
5. Reconstructing cleaned text – joining meaningful words again.

The cleaned documents were stored in a new list called cleaned_docs.

This step reduces noise and improves the accuracy of the search engine.

4. TF-IDF Model

I used TfidfVectorizer from scikit-learn to convert the cleaned documents into numerical vectors.

TF-IDF assigns higher weights to important words and lower weights to common ones.

The resulting matrix had:

- 20 documents
- 365 unique terms

I also displayed the vocabulary and converted the TF-IDF matrix into a DataFrame for easier interpretation.

5. Query Processing and Ranking

The user enters a search query.

The query goes through the same preprocessing steps as the documents:

- lowercase
- remove punctuation
- remove stopwords

Then the cleaned query is transformed into a TF-IDF vector.

Using Cosine Similarity, I calculated similarity between the query and all documents.

Finally, I sorted the documents in descending order and displayed the ranked results with similarity scores and filenames.

This completes the functionality of a simple search engine.

6. Analysis and Discussion

1) Why did the top-ranked document achieve the highest similarity score?

Because it contains the largest number of important terms that overlap with the query after preprocessing.

TF-IDF gives higher weight to meaningful words, so documents sharing these words with the query naturally score higher.

In my results, the top document included terms related to information, science, technology, which matched the query.

2) Which terms contributed most to the similarity score?

The terms that contributed most are the ones that:

- appeared in both the query and the document
- had high TF-IDF weights
- were not removed as stopwords

Examples include:

information, science, learning, systems, technology, data...

These terms represent the main topic of the query and therefore increased similarity.

3) What is the effect of removing stopwords?

Removing stopwords improved ranking accuracy because:

- stopwords appear in almost all documents
- they do not help distinguish relevant from irrelevant documents
- they add noise and reduce similarity quality

By removing them, the TF-IDF model focuses on meaningful content words, which improves the search engine's performance.

7. Conclusion

In this project, I built a functional search engine using TF-IDF and Cosine Similarity.

I uploaded 20 external documents, cleaned them, converted them into TF-IDF vectors, processed user queries, and ranked documents based on similarity. This project demonstrates the core concepts of Information Retrieval and shows how search engines convert text into numerical vectors to determine relevance.

"واخر دعواهم ان الحمد لله رب العالمين"